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Neuropsychological study in 19 French patients with White-Sutton syndrome and *POGZ* mutations

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Running head: Neuropsychological studies in White-Sutton syndrome

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Key words: Intellectual disability (ID), learning disabilities, *POGZ*, neurocognitive profile, White-Sutton syndrome

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Accepted Article

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ABSTRACT

Introduction – White-Sutton syndrome is a rare developmental disorder characterized by global developmental delay, intellectual disabilities (ID), and neurobehavioral abnormalities secondary to pathogenic *POGZ* variants. The purpose of our study was to describe the neurocognitive phenotype of an unbiased national cohort of patients with identified *POGZ* pathogenic variants.

Materials and methods – This study is based on a French collaboration through the AnDDI-Rares network, and includes 19 patients from 18 families with *POGZ* pathogenic variants. All clinical data and neuropsychological tests were collected from medical files.

Results – Among the 19 patients, 14 patients exhibited ID (6 mild, 5 moderate and 3 severe). The 5 remaining patients had learning disabilities and shared a similar neurocognitive profile, including language difficulties, dysexecutive syndrome, attention disorders, slowness and social difficulties. One patient evaluated for autism was found to have moderate autism spectrum disorder.

Conclusion – This study reveals that the cognitive phenotype of patients with *POGZ* pathogenic variants can range from learning disabilities to severe ID. It highlights that pathogenic variations in the same genes can be reported in a large spectrum of neurocognitive profiles, and that children with learning disabilities could benefit from next generation sequencing techniques.

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INTRODUCTION

In recent years, the molecular characterization of neurodevelopmental disorders has increased thanks to advances in genomic techniques [DDD Study, 2015; Gilissen *et al.,* 2014]. Exome sequencing (ES) has revealed that pathogenic variants in the same gene may be associated with a broad spectrum of neurobehavioral and neurodevelopmental phenotypes. White-Sutton syndrome (OMIM 616364) is a rare developmental disorder that is caused by a heterozygous pathogenic variants in *POGZ*. This disorder was identified in 2016, with *de novo* variants most commonly reported in the literature [White *et al.,* 2016]. The main features observed in patients with White-Sutton syndrome are developmental delay, intellectual disabilities (ID), and neurobehavioral abnormalities (including autism spectrum disorder) [White *et al.,* 2016; Stessman *et al.,* 2016; Ye *et al.*, 2015; Batzir *et al.*, 2019]. Additional features may include hypotonia, sensorineural hearing impairment, visual defects, a tendency to be overweight, gastrointestinal difficulties and facial features [White *et al.,* Stessman *et al.,* 2016, Batzir *et al.*, 2019].

POGZ (pogo transposable element-derived protein with zinc finger domain) encodes a heterochromatin protein 1 a-binding protein containing a cluster of multiple C2H2-type zinc fingers, a centromere protein (CENP) B-like DNA-binding domain, and a DDE domain that might regulate gene expression [Matsumura *et al.*, 2016; Nozawa *et al.*, 2010]. This protein is involved in mitosis and is expressed in the human brain from the fetal stage into adulthood [Stessman *et al.*, 2016].

POGZ is described as an ID gene. In the literature, *POGZ* pathogenic variants are found in patients with low-normal intelligence to severe ID. Most individuals have been described with mild to moderate ID [Stessman *et al.*, 2016 ; Batzir *et al.*, 2019], but very few details about their cognitive profile are reported. Only a few patients have undergone psychometric testing to evaluate the range of intellectual efficiency.

Following the experience of three patients with a *POGZ* pathogenic variant who were found to have learning disabilities rather than ID after completing a series of psychometric tests, we collected data from all the psychometric tests performed in individuals in the French *POGZ* cohort, which includes 19

individuals diagnosed through different next generation sequencing approaches. The purpose of our study was to describe the cognitive phenotype of individuals with identified *POGZ* pathogenic variations.

PATIENTS AND METHODS

Patient recruitment, clinical and molecular study

Patients were recruited out thanks to collaboration within the French national AnDDI-Rares network. Clinical and molecular data from 19 patients (6 males, 13 females) with White-Sutton syndrome and *POGZ* pathogenic or probably pathogenic variants were collected from medical files by the referring physician. The families agreed to data sharing.

Neuropsychological assessment

The results of neuropsychological tests were collected by the referring physician at the time of recruitment. Each patient had already been seen by a neuropsychologist or a clinical psychologist for observation and the assessment of his/her cognitive abilities, his/her behavior and/or adaptive skills. All assessments were carried out by the practitioner during the session. At the end of the assessments, the diagnosis of intellectual disabilities or learning disabilities and its level were made by the practitioner. The different scales used are summarized in Table 1.

The severity of intellectual disabilities was determined on the TIQ score. TIQ scores are commonly sorted in the following ranges: mild ID (50-69), moderate ID (35-49), severe ID (20-34), profound ID (less than 19). The calculations of the TIQ are based on several indexes: Verbal Comprehension Index, Visual Spatial Index, Fluid Reasoning Index, Working Memory Index, Processing Speed Index. All indexes need to be homogeneous for the TIQ to be representative of the overall cognitive. Faced with patients with heterogeneous indexes, the practitioner takes into account the cognitive and adaptative abilities in order to define a level of intellectual disabilities.

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The cognitive abilities of the patients were assessed either with psychometric tests or by means of a developmental scale taking into account the patient's age and their capabilities during testing. Patient 17 also underwent testing for autism with the Autism Diagnostic Observation Schedule 2nd Edition (ADOS-2), and Autism Diagnostic Interview-Revised (ADI-R).

Given the absence of ID, in order to assess their attentional and executive neurocognitive skills more accurately, patients 1, 2, 3 and 19 took additional tests including several neuropsychological test batteries: Children's Memory Scale (CMS), Kaufmann Assessment Battery for Children 2nd Edition (KABC-II), Test of Attentional Performance for Children 2nd Edition (kiTAP), Developmental NEuroPSYchological Assessment 2nd Edition (NEPSY-II), and Test of Everyday Attention for Children (TEA-Ch).

For the adaptive assessment, the Vineland Adaptive Behaviour Scale 2nd Edition (Vineland-II) was used as a family-completed questionnaire for patients 2, 3, 4, 6, 8, 11, 14, 15 and 18, and semi-directed interviews were conducted by the practitioner for patients 1, 5, 6 and 16.

RESULTS

Molecular study

In our cohort, we observed 17 *POGZ* variants, including 7 frameshift, 4 nonsense, 2 splice and 4 missense variants (Figure 1). 12 variants were undescribed, and all were *de novo* except for one family with autosomal dominant inheritance.

In this family (herein identified as Family 3) an intronic variant (NM_015100.3:c.1524-3C>G, corresponding to the acceptor splice site of exon 10) was firstly identified in Patient 2 and the subsequent segregation analysis within the family revealed that it was inherited from the affected father, where it occurred *de novo*. The father also transmitted this variant to a second child, the affected sister of Patient 2 (Patient 3). Human Splice Finder [Desmet *et al*, 2009 - http://www.umd.be/HSF/] and SPiP [Leman *et al.*, in preparation - https://github.com/spip/SPIP] analyses suggested a potential effect on splicing. HSF and SpiP predicted the creation of a new site (+

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62.51% and +66.21% with Wald interval [58.10%-81.58%] of variation respectively) that would theoretically cause the retention of 2 intronic nucleotides in parallel with a weaker activity of the canonical site (- 11.78% and -15.12% of variation respectively, Supplemental Figure 1). If confirmed, we would thus expect degradation of the transcript with the 2 nt intronic retention by the mechanism of nonsense-mediated mRNA decay (NMD). To investigate this hypothesis, we performed retrotranscription of RNA purified from blood samples, collected in PaxGene tubes, obtained from Patient 3 and her father and amplified a cDNA region encompassing exons 6 through 13. Deep targeted cDNA sequencing (MiSeq analysis) in these samples revealed that 12-15% of the transcripts showed the expected 2 nt retention as well as a small proportion (roughly 1%) of exon skipping events. Both splicing anomalies were identified in Patient 3 and her father and were absent in unrelated healthy individuals (Supplemental Figure 2). These evidences were in favour of a NMD mechanism, however quantitative RT-PCR analysis could not confirm this hypothesis as overall levels of *POGZ* expression in Patient 3 and her father were almost indistinguishable from those detected in control samples (data not shown), indicating that quantification of *POGZ* expression in blood may not be informative perhaps due to the interindividual variability and the relevance of the tissue analysed.

Patient 19 also was also carrier of a microdeletion in 15q11.2 region, of size of 433 kb, inherited from her asymptomatic mother. This deletion was known as a predisposing factor for neurodevelopmental disorders with incomplete penetrance and variability of expression (breakpoints 1 and 2 of the Prader Willi/Angelman syndrome critical region). The proportion of this variant to symptoms of patient 19 was considered modest [Jønch *et al.*, 2019].

Clinical study

The clinical features of the 19 individuals with *POGZ* variants are summarized in Table 2. At the time of recruitment, the patients were between 7 and 21 years old, with a median age of 11.7 years. Most of the patients had speech delay. None had experienced seizures. Brain MRI, which was done in 15 patients, revealed cerebellar atrophy in patient 1, microcephaly and dysgenesis of corpus callosum in

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patient 4, vermian atrophy in patient 5, olfactive bulb hypoplasia and thin corpus callosum in patient 6 and delayed myelination in patient 7. The MRI of the 10 remaining patients were interpreted as normal. Gastro-intestinal manifestations were reported in some patients: oral sensitivity disorders (3/13), constipation (4/13), occlusion (2/13) and vomiting (3/13). Some patients presented congenital malformations: genitourinary malformations (3/19), heart malformation (2/19) and diaphragmatic hernia (1/19). Half of the patients presented a tendency to overweight. Though most individuals exhibit facial features, there are no significant characteristic facial features for this disorder.

Among the 19 patients, 13 exhibited a behavioural disorder (Supplemental table 1), including anxiety (10/19), stereotypies (6/19), inappropriate behaviour (9/19), social withdrawal (7/19), and hyperactivity (2/19). Four patients were reported to have sleep difficulties.

Neuropsychological assessment

The norm is an interval of two standard deviations around the mean of the scores obtained by the reference population. It takes inter-individual variations into account and indicates the expected functioning in the assessed domain. The results of each patient for each test were compared to the norm for people of their age at the time of the assessment (Table 3A and 3taB).

The five patients with learning disabilities (patient 1, 2, 3, 9 and 19) obtained heterogeneous results that did not allow for the calculation of a TIQ (Total Intellectual Quotient) that could faithfully represent their overall cognitive profile. The weaknesses of patients 1, 2, 3 and 9 were mainly their verbal working memory and processing speed. These results suggested attention and/or executive dysfunctions.

Patients 1, 2, 3 and 19 were assessed for attention and executive functioning (Table 3B). Their weaknesses were mainly sustained attention, divided attention, planning and cognitive flexibility. This confirmed attention and executive deficits. Patients 1, 2 and 3 were also assessed for social skills, revealing weaknesses mainly in the field of theory of mind. Adaptive skills were assessed in these three

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patients. Patients 2 and 3 obtained heterogeneous results that did not allow for the calculation of an overall adaptive behavioural score.

Patients 2 and 3 were found to have the most difficulty with communication. This can be explained by verbal and/or attention and/or executive and/or social cognition dysfunctions. Patient 1 had a global adaptive profile below the norm. It is worth noting that all the patients with learning disabilities in this study presented language difficulties even though verbal comprehension did not always emerge as a weakness on the IQ scale.

The neurocognitive, social cognition and adaptive profiles of the five patients with learning disabilities was heterogeneous (Figure 2). Several areas were consistently found to be weak for these patients: verbal working memory, speed of treatment, sustained attention, divided attention, planning and cognitive flexibility (neurocognition), the theory of mind (social cognition) and communication (adaptive behaviour). These findings were suggestive of the language, attention, executive and social dysfunctions that affected the patients' daily lives.

Among the patients with mild ID, three patients (patients 10, 13 and 16) obtained heterogeneous cognitive results, and their TIQs could therefore not be calculated. Similar to the patients with learning disabilities, verbal working memory seemed to be a weakness for two patients (patients 10 and 16) as well as the speed of treatment (patients 10 and 13). Two other patients (patients 14 and 16) obtained heterogeneous results, so the adaptive behaviour score could not be calculated. Similar to the patients with learning disabilities, communication skills were a weakness for these two patients.

Overall, the patients with moderate to severe ID appeared to have relatively homogeneous cognitive profiles and adaptive functioning that did not demonstrate specific strong and/or weak points with the scales used (pc <0,1). Only one patient with a moderate ID (patient 11) presented non-homogeneous adaptive behaviour with weakness in communication, similar to the patients with learning disabilities. In patient 7, the results of the Brunet-Lezine scale indicated a global developmental delay. In patient 17, the PEP-3 test found a global developmental delay with particular weaknesses in expressive

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language and motor skills. Calculation of IQT was not possible with these scales. The autism evaluation scale was only tested in patient 17, with results indicating moderate autism spectrum disorder.

Similar behavioural characteristics seemed to be found for all the tested patients regardless of their cognitive diagnosis. Clinical observation during the assessments showed immature behaviour (n=7), significant slowness (n=7) and attention disorders (n=8). These were associated with either excessive inhibition leading to withdrawal, passivity and lack of initiative (n=3) or a lack of inhibition leading to agitation, opposition or even provocation (n=4). An anxious temperament was also found in patients with heterogeneous cognitive profiles (learning disabilities and mild ID, n=7), which did not seem to be shared by subjects with homogeneous mild, moderate or severe ID.

These qualitative data were consistent with the quantitative results obtained from the evaluation of patients with learning disabilities, specifically dysfunction related to language, attention, executive and social skills.

DISCUSSION

Our study revealed that patients with White-Sutton syndrome have broadly heterogeneous global cognitive profiles. They exhibited levels of intellectual efficiency ranging from learning disability to severe intellectual disability. Fourteen patients were found to have ID (6 mild, 5 moderate and 3 severe). The five remaining patients had learning disabilities and presented heterogeneous cognitive profiles with language, attention, executive and social dysfunctions that affected their daily lives (especially communication). Their weaknesses were essentially associated with difficulties in verbal working memory, processing speed, sustained and divided attention, planning and cognitive flexibility, theory of mind, and communication.

Three of the patients with mild ID presented heterogeneous cognitive profiles. Two of them and one of the patients with moderate ID presented heterogeneous adaptive profiles. The cognitive and adaptive weaknesses were similar in patients with learning disabilities. No recognizable cognitive profile was found otherwise among the patients carrying *POGZ* pathogenic variant with ID.

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A variety of evaluations were used to assess the patients in this study. The assessments were different depending on the age of the patient and their capabilities at the time of the test, and they were carried out by different practitioners. A prospective study using the same assessment tools at a same age and with the same practitioner for each patient would provide more coherent analyses.

In addition, while the scales used for the diagnosis of ID are suitable to establish a profile with strengths and weaknesses in patients with mild ID, these scales are less suitable for patients with moderate and severe ID. Indeed, these patients often obtain minimum scores rendering their profile homogeneous thus preventing a fine assessment of their strengths and weaknesses.

In the literature, the cognitive profiles of patients with White-Sutton syndrome was usually only briefly discussed. The patients in the study by White *et al.* [2016] were 5 years in average, which explains the calculation of TIQ for only one 15-year-old patient. Developmental assessments had been carried out for 3 out of the 4 remaining patients. In Stessman *et al.'s* [2016] large cohort of 25 individuals, psychometric tests are mentioned in the supplementary data for 10 patients and developmental assessments for 3 patients. Among the patient who were assessed with psychometric tests, three presented learning disabilities and seven mild ID. For the 12 remaining patients, it was not possible to determine whether the range of intellectual ability was obtained by psychometric tests or by the clinical impression of the referring physician. Recently, Batzir *et al.* [2019] described 22 individuals carrying a *POGZ* pathogenic variant including 19 patients with learning difficulties, 13 patients of which had ID. The range of intellectual ability was not specified. The number of patients who performed psychometric tests in the literature and in the present study are indicated in Figure 3A, including their range of ID. Half of all patients presented mild ID, 14% moderate ID and 26 % had learning disabilities. Severe ID has been diagnosed in patients who obtained negative results on the adaptive scale (Vineland-II) and were unable to pass the psychometric tests. Figure 3B summarizes the range of ID obtained with psychometric tests or by the clinical impression of the referring physician for the patients in the literature and in the present study. 17% of the patients presented learning disabilities, 36% presented mild ID, 17% presented moderate ID and a 32% of patients had severe ID.

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In view of the results, it seems important to objectively evaluate the level of intellectual efficiency of these patients using psychometric tests. For instance, of patients 2 and 3 the clinical impression of the referring physician was "ID". However, the neuropsychological test diagnosed low-average and borderline intellectual efficiency. This confirms the importance of clarifying the cognitive profile of these patients in order to provide suitable education and career guidance.

For instance, patient 11 was described as having a moderate ID based on a cognitive evaluation performed before age 4, which is young to make such a diagnosis. This patient is now 10 years old and is the only moderate ID patient with above minimum of adaptive behaviour scores (ranging from low to average), indicating that his skills are worth reassessing.

It is commonly understood that cognitive deficits in patients with ID develop during developmental years at individually varying rate regardless of the etiology. Furthermore, the course of cognition could vary depending on the syndrome [Sauna-aho *et al.*, 2019]. In our study, patients were assessed at different ages. Three patients were adults with intellectual efficiency ranging from mild to moderate ID, and 7 patients were more than 10 years old with intellectual efficiency ranging from learning disabilities to severe ID. To evaluate the course of cognition in patients with White-Sutton, it would be interesting to follow-up these patients in adulthood.

Nevertheless, this study confirms the broad heterogeneity of intellectual efficiency in patients with White-Sutton syndrome, ranging from learning disabilities to severe ID [Stessman *et al.,* 2016]. This suggests that the same genes could be responsible for a large spectrum of neurocognitive profiles. In view of our findings, the use of next generation sequencing techniques such as exome sequencing in children with learning disabilities is worth discussing. These patients could greatly benefit from genetic diagnosis in order to provide suitable education and career guidance.

In addition to the results of the neuropsychological assessments, the clinical symptoms of our patients were similar to what has been reported in previous studies. Most patients had global developmental delay. Language development was more affected than motor development. One patient received a diagnosis of autism spectrum disorder, while most of the others presented behavioural disorders.

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Among the 15 patients who underwent brain MRI, 5 patients were found to have central nervous system (CNS) malformations (33%). This proportion was similar to the proportion found in the literature review by Dentici *et al.* in 2017. CNS malformations appeared to represent a recurrent feature of patients with *POGZ* pathogenic or likely pathogenic variant.

In our study, there was no obvious genotype-phenotype correlation. Indeed, splice variants and missense variants were found in patients with learning disabilities as well as in patients with intellectual disabilities. But the number of patients may be too small to raise conclusions.

In conclusion, our study contributed to current knowledge regarding the cognitive profile of patients with White-Sutton syndrome. This was the first study to focus on the neurocognitive profile of these patients, which ranged from learning disabilities to severe ID. We also point out that the same genes can be reported in a large spectrum of neurocognitive profiles, and that children with learning disabilities could benefit from next generation sequencing techniques such as exome sequencing.

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LEGENDS TO FIGURES AND TABLES

Figure 1. Protein model of POGZ with identified variants indicated. All mutations (indicated by individual identifiers that correspond to Table 2) have been annotated on the RefSeq transcript (GenBank: NM_015100.3) (POGZ). Protein domains are indicated on the structure. Abbreviations are as follows: ZNF, zinc finger; HTH, helix-turn-helix; del, deletion; fs, frameshift; *, premature stop codon.

Figure 2. Graphical representation of psychometric tests performed by patients with learning disabilities. Above the full line represents the average, between the full line and the dotted line represents the borderline range, below the full line represent the deficit range.

Figure 3A. Distribution of intellectual disabilities after psychometric assessment in patients with POGZ variant. Abbreviations are as follows: LD, learning disabilities; ID, intellectual disabilities.

Figure 3B. Distribution of intellectual disabilities after psychometric assessment or clinical impression of the referring physician in patients with POGZ variant. Abbreviations are as follows: LD, learning disabilities; ID, intellectual disabilities.

Table 1. The different tests performed by each patient. Abbreviations are as follows: KABC-II, Kaufman Assessment Battery for Children 2nd Edition; CMS, Children's Memory Scale; NEPSY-II, Developmental NEuroPSYchological Assessment 2nd Edition; TEA-Ch, Test of Everyday Attention for Children; kiTAP, Test of Attentional Performance for Children 2nd Edition ;WISC-V, Wechsler Intelligence Scale for Children 5th Edition; WAIS-IV, Wechsler Adult Intelligence Scale 4th Edition; BLR, Psychomotor development of early childhood Scale- Brunet Lezine; WPPSI-IV, Wechsler Preschool and Primary Scale of Intelligence 4th Edition; WPPSI-III, Wechsler Preschool and Primary Scale of Intelligence 3rd Edition; WNV, Wechsler Nonverbal Scale of Ability; PEP-3, Psychoeducational Profile, 3rd Edition; ADOS-2,

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Autism Diagnostic Observation Schedule 2nd Edition; ADI-R, Autism Diagnostic Interview-Revised; Vineland-II, Vineland Adaptive Behavior Scale 2nd Edition.

Table 2. Clinical features and molecular data of individuals harbouring *POGZ* variant. All HGVS annotations were annotated on RefSeq transcript NM_015100.3. Abbreviations are as follows: M, male; F, female; +, present; -, not present; ND, no data; del, deletion; fs, frameshift; *, premature stop codon; LD, learning disabilities.

Table 3A. Psychometric and adaptive tests. Patients 7, 12 and 17 did not perform psychometric and adaptive tests. Abbreviations are as follows: LDs, learning disabilities; ID, intellectual disability; NT, not tested; HTG, heterogeneous

Table 3B. Additional tests (from CMS, KABC-II, NEPSY-II and TEA-Ch) and clinical observation in patients 1, 2, 3 and 19

Supplemental Figure 1. Graphical representation of SPiP analysis of the c.1524-3C>G variant. Schematic representation of a portion of the *POGZ* gene. The red circle at exon 10 indicates the genomic position of the acceptor splice site containing the variant (A). The c.1524-3C>G variant is predicted to weaken the reference acceptor site (score= 0.1546 and 0.0034 in wild-type and mutated allele respectively) and create a new site (score= 0.0263), causing the retention of two intronic nucleotides (B).

Supplemental Figure 2. Targeted amplicon-sequencing of retrotranscribed *POGZ* mRNAs derived from blood samples of Family 3 and unaffected individuals (CTRLs). Sashimi plot indicating the mean and the proportion of aberrant splicing events, including exon skipping and intronic retention occurring in affected individuals of Family 3 (Patient 3 and father of the proband) as compared to unaffected and unrelated individual. (**A**). Inset of exon 10 showing the inclusion of two intronic nucleotides at the acceptor splice site containing the heterozygous c.1524-3C>G variant. (**B**).

Supplemental Table 1. Behavioural disorders of individuals harbouring a *POGZ* variant

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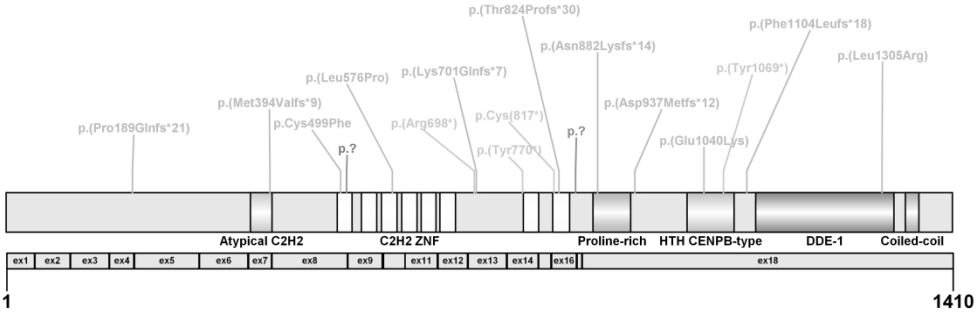

Figure 1. Protein model of POGZ with identified variants indicated. All mutations (indicated by individual identifiers that correspond to Table 2) have been annotated on the RefSeq transcript (GenBank: NM_015100.3) (POGZ). Protein domains are indicated on the structure. Abbreviations are as follows: ZNF, zinc finger; HTH, helix-turn-helix; del, deletion; fs, frameshift; *, premature stop codon.

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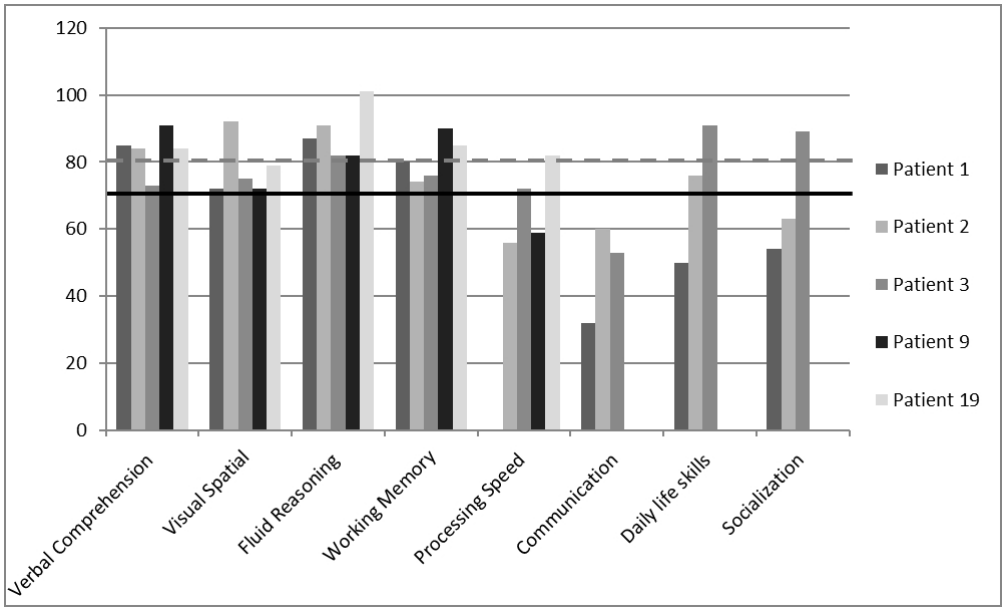

Figure 2. Graphical representation of psychometric tests performed by patients with learning disabilities. Above the full line represents the average, between the full line and the dotted line represents the borderline range, below the full line represent the deficit range.

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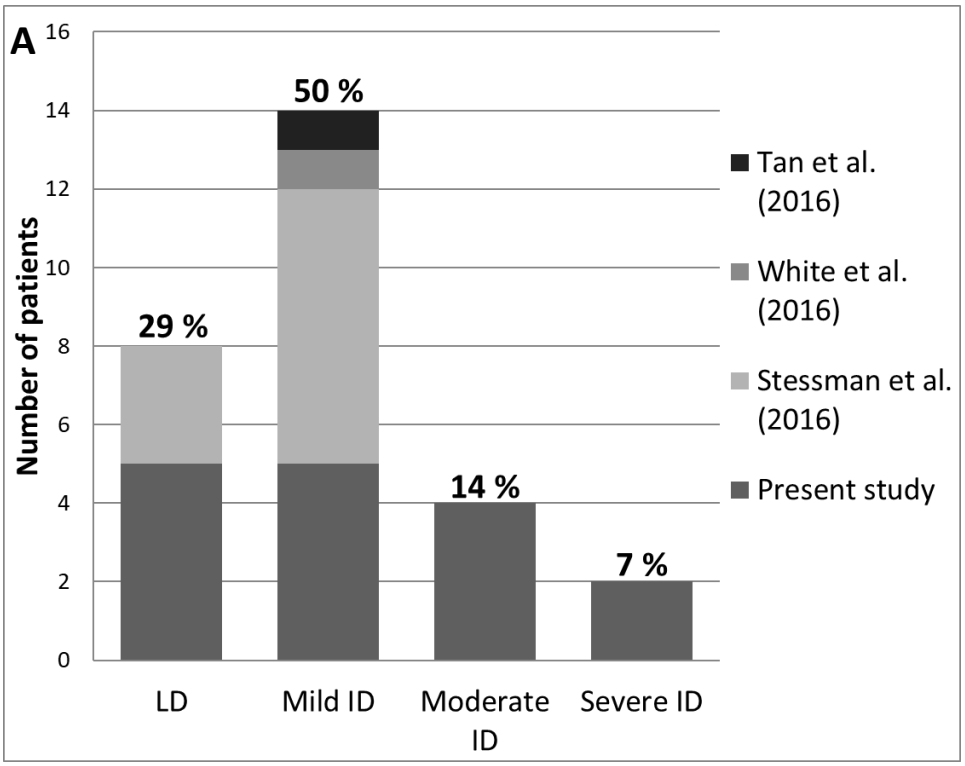

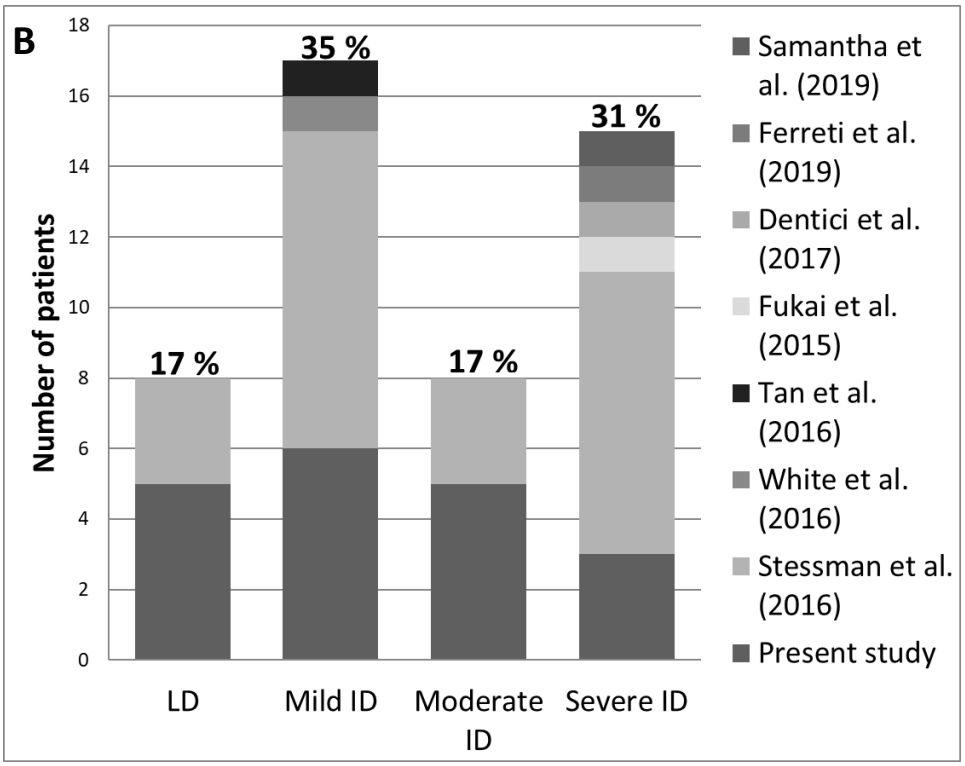

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Table 1. The different tests performed by each patient

	Developmental and cognitive assessment		Adaptative assessment	
	Test	Age at the assessment (years ; months)	Test	Age at the assessment (years ; months)
Patient 1	KABC-II	10 ; 1	Vineland-II interview	12 ; 10
	CMS, NEPSY-II, TEA-Ch	12 ; 5		
Patient 2	WISC-V, CMS, kiTAP, NEPSY-II, TEA-Ch	8 ; 6	Vineland-II questionnaire	8 ; 6
Patient 3	WISC-V, NEPSY-II, TEA-Ch	10 ; 9	Vineland-II questionnaire	10 ; 9
Patient 4	WAIS-IV	19 ; 6	Vineland-II questionnaire	19 ; 6
Patient 5	-	-	Vineland-II interview	15 ; 4
Patient 6	KABC-II	13 ; 5	Vineland-II interview	13 ; 5
			Vineland-II questionnaire	18 ; 1
Patient 7	BLR	4 ; 11	-	-
Patient 8	-	-	Vineland-II questionnaire	8 ; 0
Patient 9	WPPSI-IV	4 ; 2	-	-
Patient 10	WISC-V	14 ; 11	-	-
Patient 11	WPPSI-III NEPSY-II	3 ; 11	Vineland-II questionnaire	10 ; 2
Patient 12	-	-	-	-
Patient 13	WISC-V	14 ; 9	-	-
Patient 14	WNV, WPPSI-IV	6 ; 8	Vineland-II questionnaire	6 ; 8
Patient 15	WAIS-IV	21 ; 5	Vineland-II questionnaire	21 ; 5
Patient 16	WISC-V	14 ; 0	Vineland-II interview	13 ; 9
Patient 17	ADOS-2, ADI-R	3 ; 4	-	-
	PEP-3	4 ; 10 – 6 ; 0		
Patient 18	WAIS-IV	19 ; 5	Vineland-II questionnaire	19 ; 5
Patient 19	WPPSI-IV, CMS, KABC-II, kiTAP, NEPSY-II, TEA-Ch	6 ; 4	-	-

Abbreviations are as follows: KABC-II, Kaufman Assessment Battery for Children 2nd Edition; CMS, Children's Memory Scale; NEPSY-II, Developmental EuroPSYchological Assessment 2nd Edition; TEA-Ch, Test of Everyday Attention for Children; kiTAP, Test of Attentional Performance for Children 2nd Edition ;WISC-V, Wechsler Intelligence Scale for Children 5th Edition; WAIS-IV, Wechsler Adult Intelligence Scale 4th Edition; BLR, Psychomotor development of early childhood Scale- Brunet Lezine; WPPSI-IV, Wechsler Preschool and Primary Scale of Intelligence 4th Edition; WPPSI-III, Wechsler Preschool and Primary Scale of Intelligence 3rd Edition; WNV, Wechsler Nonverbal Scale of Ability; PEP-3, Psychoeducational Profile, 3rd Edition; ADOS-2, Autism Diagnostic Observation Schedule 2nd Edition; ADI-R, Autism Diagnostic Interview-Revised; Vineland-II, Vineland Adaptive Behavior Scale 2nd Edition.

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Table 2. Clinical features and molecular data of individuals harbouring *POGZ* variant

	Patient 1	Patient 2	Patient 3	Patient 4	Patient 5	Patient 6	Patient 7	Patient 8	Patient 9
Variation	c.3206_3207delAT p.Tyr1069*	c.1524-3C>G p. ?	c.1524-3C>G p.?	c.2451C>A p.Cys817*	c.2451C>A p.Asn882Lysfs*14	c.2400dup p.Lys701Glnfs*7	c.2836delG p.Asp937Metfs*12	c.2545+1delG p. ?	c.566del p.Pro189Glnfs*21
Inheritance	De novo	Inherited from the father (affected)	Inherited from the father (affected)	De novo	De novo	De novo	De novo	De novo	De novo
Age	12	9	11	19	15	17	9	7	8
Gender	F	M	F	F	F	F	F	M	F
Intellectual disability	LD	LD	LD	Moderate	Severe	Mild	Severe	Severe	LD
First words / Sentences	12 months / 5 years	3 years / ND	18 months / ND	3 years / ND	3 years 3 months	20 months / 5 years	No speech	No speech	24 months / ND
Walking age	18 months	18 months	14 months	24 months	2 years 10 months	17 months	2 years 5 months	6 years	21 months
Hearing loss	-	-	-	ND	-	+	-	+	-
Vision problems	-	+	+	ND	-	-	+	+	+
Seizures	-	-	-	-	-	-	-	-	-
Cerebral MRI	Cerebellar atrophia	ND	ND	Microcephaly, dysgenesis of the corpus callosum	Vermian atrophy	Normal	Olfactive bulbs hypoplasia, thin corpus callosum	Delayed myelination	Normal
Gastro-intestinal manifestations	Oral sensory disorders, constipation	ND	ND	ND	Vomiting	ND	Constipation	Oral sensory disorders constipation, occlusion	-
Congenital malformations	-	-	-	-	Left megauretere, right ureteral duplication with bilateral reflux	-	Double renal sinus	Common mesentary, atrial septal defect, micropenis, cryptorchidism	-
Hypotonia	+	ND	ND	ND	ND	+	+	ND	+
Tendance to overweight	+	+	+	+	ND	+	ND	ND	ND
Morphological features	Wide nasal bridge, drooping cheeks, frozen smile, few expressive face, bradymetacarpia and flat feet	No particular morphological features	Synophrys, prominent nasal bridge, microcephaly	High-arched palate, abdominal café-au-lait spot, elbow extension limitation with ulna valga, right gibbosity	Hypertelorism, short palpebral fissures, pear-shaped nose	No particular morphological features except high arched palate	Prognathism, everted vermillon of the upper lip, square-shaped face	Round face, telecanthus, epicanthus inversus, upslanted palpebral fissures, short nose, anteverted nostrils, downturned corners of the mouth, brachycephaly	Brachycephaly, round face, hypertelorism, upslanted palpebral fissures, up-turned nose, long/large philtrum, oro-facial syndrome with macroglossia, hypersalivation, open mouth, prognathism

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Table 2. Clinical features and molecular data of individuals harbouring *POGZ* variant *(Continued)*

Patient 10	Patient 11	Patient 12	Patient 13	Patient 14	Patient 15	Patient 16	Patient 17	Patient 18	Patient 19
c.1727T>C p.Leu576Pro	c.3914T>G p.Leu1305Arg	c.3312del p.Phe1104Leufs*18	c.2092C>T p.Arg698*	c.3118G>A p.Glu1040Lys	c.1180_1181del p.Met394Valfs*9	c.1180_1181del p.Met394Valfs*9	c.2310C>G p.Tyr770*	c.2469del p.Thr824Profs*30	c.1496G>T p.Cys499Phe
De novo	De novo	De novo	De novo	De novo	De novo	De novo	De novo	De novo	De novo
16	9	7	15	7	21	14	7	20	6
M	F	M	F	F	M	F	M	F	F
Mild	Mild	Mild	Mild	Mild	Mild	Mild	Moderate	Moderate	LD
ND / 6 years	ND	2 years / ND	30 months / ND	ND	18 months / ND	4 years / ND	36 months	18 months	Before 3 years / ND
21 months	18 months	17 months	18 months	14 months	15 months	21 months	30 months	12 months	18 months
-	-	-	-	ND	+	+	-	-	-
+	-	+	-	+	+	+	+	-	+
-	-	-	-	-	-	-	-	-	-
Normal	ND	Normal	Normal	Normal	Normal	Normal	Normal	Normal	ND
Oral sensory disorders	ND	Cyclic vomiting	ND	Constipation, occlusion	ND	-	-	Cyclic vomiting	-
-	-	-	-	-	Diaphragmatic hernia	Patent foramen ovale	-	-	-
+	-	-	+	-	-	-	-	-	+
+	ND	+	ND	+	+	+	-	+	+
Right preauricular tag, columella hypoplasia, long philtrum, everted vermillion of the lower lip	Hypertelorism, low set ears, thin upper lip, protruding columella, wide central upper incisors	Microcephaly, brachycephaly, hypotonic facies, flat nasal bridge, overfield ears	High forehead, hypoplasia of the middle face, slight hypertelorism	Synophrys, hypertrophy of the nasal bone, thin lips, dental diastasis, tapering fingers, café-au-lait spots, angioma	Thin upper lip, brachycephaly, epicanthal folds	Large mouth, epicanthus, hypertelorism, prognathism	No particular morphological features, small mouth	Round and hypotonic face, downslanted palpebral fissures, midface hypoplasia, short philtrum, thin upper lip, everted lower lip, anteverted ears, dried and thick hair	No particular morphological features, bitemporal retraction

All HGVS annotations were annotated on RefSeq transcript NM_015100.3. Abbreviations are as follows: M, male; F, female; +, present; -, not present; ND, no data; del, deletion; fs, frameshift; *, premature stop codon; LD, learning disabilities.

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Table 3A. Psychometric and adaptive tests

	Patient 1	Patient 2	Patient 3	Patient 4	Patient 5	Patient 6	Patient 8	Patient 9	Patient 10	Patient 11	Patient 13	Patient 14	Patient 15	Patient 16	Patient 18	Patient 19
Conclusion	LD	LD	LD	Moderate ID	Severe ID	Moderate ID	Severe ID	LD	Mild ID	Moderate ID	Mild ID	Mild ID	Mild ID	Mild ID	Moderate ID	LD
Total Intellectual Quotient	HTG	HTG	HTG	Deficit	NT	Deficit	NT	HTG	HTG	Deficit	HTG	Deficit	Deficit	HTG	Deficit	HTG
Verbal Comprehension Index	Low average	Low average	Borderline	Deficit	NT	NT	NT	Normal	Deficit	Deficit	Deficit	Deficit	Deficit	Deficit	Deficit	Low average
Visual Spatial Index	Borderline	Normal	Borderline	Deficit	NT	Deficit	NT	Borderline	Borderline	Deficit	Borderline	Deficit	Deficit	Low average	Deficit	Borderline
Fluid Reasoning Index	Low average	Normal	Low average	Deficit	NT	NT	NT	Low average	Deficit	NT	Deficit	Borderline	Deficit	Borderline	Deficit	Normal
Working Memory Index	Low average	Borderline	Borderline	Deficit	NT	Deficit	NT	Normal	Deficit	NT	Normal	Deficit	Deficit	Deficit	Deficit	Low average
Processing Speed Index	NT	Deficit	Borderline	Deficit	NT	NT	NT	Deficit	Deficit	NT	Deficit	Borderline	Deficit	Borderline	Deficit	Low average
Adaptive behaviour	HTG	HTG	HTG	Deficit	Deficit	Deficit	Deficit	NT	NT	HTG	NT	HTG	Low average	HTG	Deficit	NT
Communication	Deficit	Deficit	Deficit	Deficit	Deficit	Deficit	Deficit	NT	NT	Deficit	NT	Borderline	Low average	Deficit	Deficit	NT
Daily life skills	Deficit	Borderline	Normal	Deficit	Deficit	Deficit	Deficit	NT	NT	Low average	NT	Normal	Low average	Normal	Deficit	NT
Socialization	Deficit	Deficit	Low average	Deficit	Deficit	Deficit	Deficit	NT	NT	Normal	NT	Borderline	Low average	Borderline	Deficit	NT

Patients 7, 12 and 17 did not perform psychometric and adaptive tests.
Abbreviations are as follows: LDs, learning disabilities; ID, intellectual disability; NT, not tested; HTG, heterogeneous

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Table 3B. Additional tests (from CMS, KABC-II, NEPSY-II and TEA-Ch) and clinical observation in patients 1, 2, 3 and 19

	Patient 1	Patient 2	Patient 3	Patient 19
Selective attention	Deficit	Normal	Normal	Normal
Sustained attention	Deficit	Deficit	Deficit	Deficit
Divided attention	Deficit	Deficit	Deficit	Deficit
Planning	Deficit	Deficit	Deficit	Deficit
Inhibition	Deficit	Normal	Normal	Deficit
Flexibility	Deficit	Deficit	Deficit	Deficit
Affect recognition	Low average	Deficit	Low average	NT
Theory of mind	Deficit	Deficit	Deficit	NT

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